



BUSINESS ANALYSIS

Lecture 4: Four Types of Business Analyses

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MORE ABOUT PREDICTIVE ANALYTICS

- **What is Predictive Analytics**

“Predictive analytics is the use of data to predict future trends and events. It uses historical data to forecast potential scenarios that can help drive strategic decisions.”

- Prediction of events in the near future: e.g., failure of a machinery
- Prediction of events in the distant future: e.g., company’s financial prospects in the year next to the current one

- **Methods of Predictive Analytics: Examples**

- Regression analysis
- Probabilities (try to capture uncertainty)
 - Risk analysis
 - Forecasting

Harvard Business School; What is predictive analytics: 5 examples; <https://online.hbs.edu/blog/post/predictive-analytics>

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MORE ABOUT PREDICTIVE ANALYTICS

Examples of Predictive Analytics

- Financial forecasting
Using historical data from financial statements to forecast sales, revenue, and expenses in the next year
- Forecasting staffing needs
For example, a hospitality business can analyze data on venue use to predict staffing needs
- Sales
Using historical data on customer behavior to predict which products will sell
- Manufacturing
For example training algorithms to predict when a machine will malfunction
- Healthcare
Predicting when an allergic reaction will occur and how severe it will be

Harvard Business School; What is predictive analytics: 5 examples; <https://online.hbs.edu/blog/post/predictive-analytics>

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MORE ABOUT PRESCRIPTIVE ANALYTICS

• What is Prescriptive Analytics

“The process of using data to determine an optimal course of action. By considering all relevant factors, this type of analysis yields recommendations for next steps. Because of this, prescriptive analytics is a valuable tool for data-driven decision-making.”

• Examples of Methods of Prescriptive Analytics

- Machine Learning: e.g., using algorithms on data with “if” and “else” statements to make recommendations on different combinations of stakeholder requirements
- Manual analysis: Data-driven decision making

Harvard Business School; What is predictive analytics: 5 examples; <https://online.hbs.edu/blog/post/prescriptive-analytics>

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MORE ABOUT PRESCRIPTIVE ANALYTICS

Examples of Prescriptive Analytics

- **Investment decisions**
For example, as an entrepreneur deciding which startup or investment will yield better returns
- **Sales**
Deciding which actions by prospective customers are likely to convert into purchases based on e.g.,
 - Page views
 - Email interactions
 - Site searches
 - Content engagement, such as attending webinars, downloading e-books
- **Algorithm-based actions**
Using engagement history with websites to determine content preference and prescribe content/products accordingly
- **Fraud prevention**
Example: Algorithms detect suspicious behavior of bank customers leading to automated actions such as credit card blocking
- **Marketing**
Deciding which products to design based on customer behavior and preferences

Harvard Business School; What is predictive analytics: 5 examples; <https://online.hbs.edu/blog/post/prescriptive-analytics>

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BUSINESS ANALYTICS SKILLS

- **Data Literacy**
Familiarity with different types and sources of data and analytical tools and techniques
- **Data Collection**
Knowledge of data collection techniques such as surveys, interviews, questionnaires
- **Data Analysis**
Methods for analyzing and interpreting through math, stats, ML
- **Communication**
Communication skills for connecting with stakeholders including report writing, presentations etc.
- **Data Visualization**
Techniques for visual presentation of your data e.g., charts, graphs, videos

Harvard Business School; Business analytics skills; <https://online.hbs.edu/blog/post/business-analytics-skills>

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ESSENTIAL READINGS

Harvard Business School Online

Page Title: What is predictive analytics

Available at: <https://online.hbs.edu/blog/post/predictive-analytics>

Harvard Business School Online

Page Title: What is prescriptive analytics

Available at: <https://online.hbs.edu/blog/post/prescriptive-analytics>

Harvard Business School Online

eBook: A Beginner's Guide to Data & Analytics

Available at: <https://online.hbs.edu/blog/post/descriptive-analytics>